Toronto Metropolitan University

Metropolitan Undergraduate Engineering Society

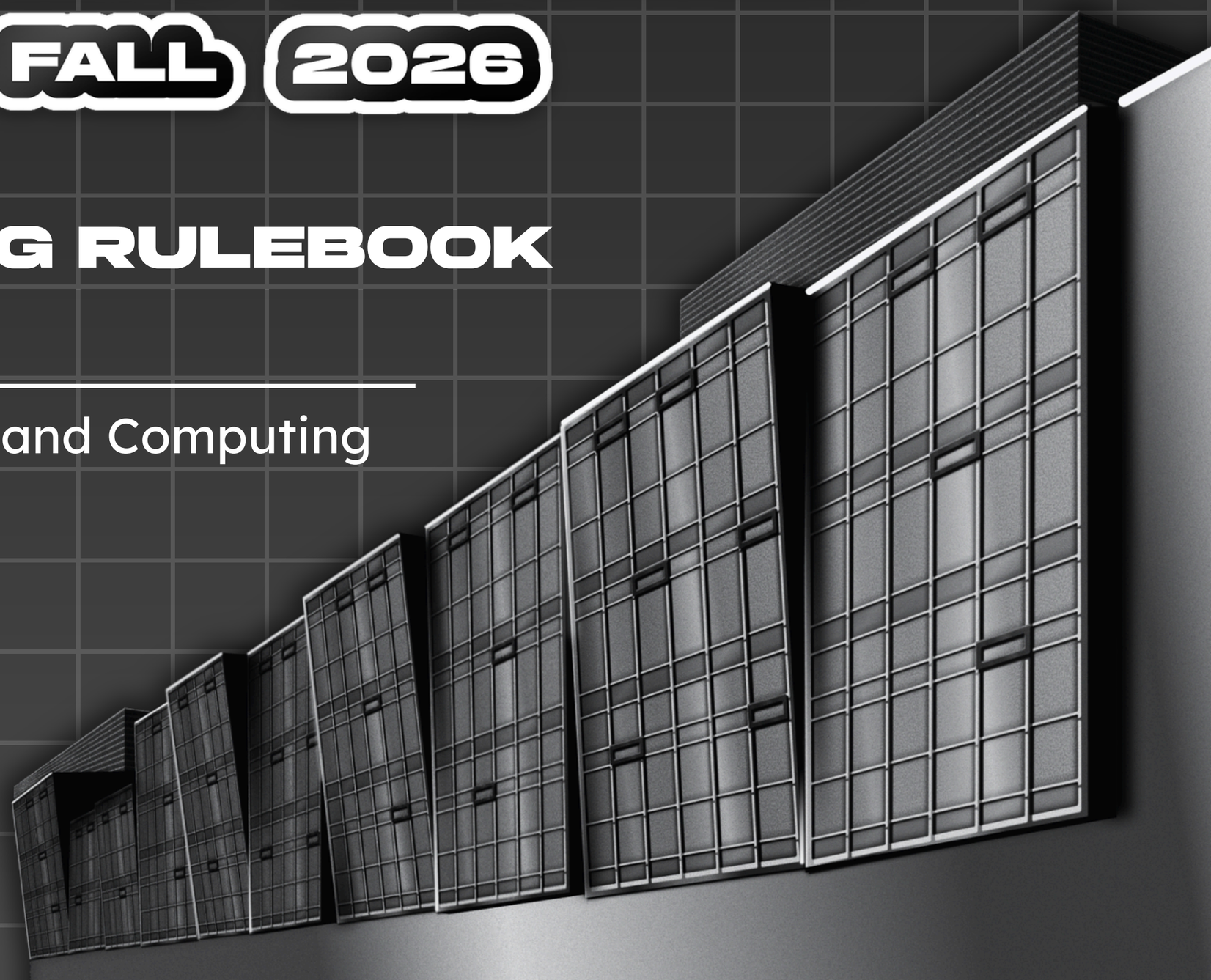
TORONTO METROPOLITAN UNIVERSITY ENGINEERING COMPETITION

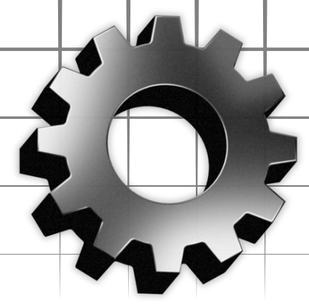
FALL 2026

PROGRAMMING RULEBOOK

NOVEMBER 7 & 8, 2026

George Vari Engineering and Computing Centre





1.1 GOAL:

The goal of the programming category is to encourage engineering students to produce a piece of readable software. Teams will use their software development, technical writing, and project management skills to design a solution to a given problem. This solution will then be presented to company executives (judging panel) for approval. The winning solution will not necessarily be the most technically correct but the one that has the most real-world application and is most thoroughly thought out.

1.2 TEAM COMPOSITION:

The Programming team will consist of a maximum of four (4) competitors.

1.3 LANGUAGE:

The presentation and any presentation materials must be done in English. However, competitors may use any language to create the program, as long as it can run and be displayed during presentations.

1.4 TOPIC:

The topic will be a real-life problem found in any professional industry which can be solved through the application of programming. The types of industries can include, but are not limited to, finance, health, transportation, manufacturing and construction. Although not completely necessary, a team that is formed of students from more than one engineering discipline is advised, as it would help to develop a complete solution.

1.5 RESOURCES:

1.5.1 Facilities Required:

- One (1) amphitheatre (presentation of problem and team presentations).
- One (1) workroom per team (design of solution).

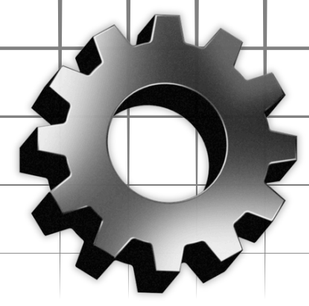
1.5.2 Personnel Required:

1.5.2.1 Judges:

A minimum of three (3) judges (and in excess, an odd number of judges) are required to assess the problem-solving abilities, proposed solution, communication skills and team dynamics of the competitors. Judges in this category should come from a variety of backgrounds, including communications, sales and technical or software engineering experience related to the topic.

1.5.2.2 Competition Leads:

The Competition Leads are responsible, along with the VP Competitions, for the design and implementation of the Programming competition. The Competition Leads must present the design problem at the beginning of the competition and answer any questions raised by competitors. Only the Competition Leads may answer questions during the design period. The Competition Leads will also answer any questions the judges may have during the competition.



1.5.2.3 Timekeeper:

The Official Timekeeper must be responsible for enforcing time limits during the question period, solution development and presentations, as stated in the rules.

1.5.3 Equipment Provided By MEC:

The following equipment will be made available to all competitors during the competition time period:

1.5.3.1 During the design phase:

- A design room with at least one (1) table, four (4) chairs and a whiteboard or blackboard.
- Internet connectivity.
- Information relevant to the design problem, at the discretion of the Competition Leads.

1.5.3.2 During the presentation:

- One (1) digital projector.
- One (1) computer containing the team’s presentation file.
- Whiteboard(s) or blackboard(s).

1.5.4 External Resources Allowed:

The competitors are allowed to use the following equipment:

- Any background research conducted by team members before the competition.
- Any electronic material stored on CDs, USB keys, disks or other storage devices.
    - Please make sure with the Programming Coordinator that the format of any electronic information will be accessible using the equipment used by MEC.
- Any textbooks, course notes or other reference material.
- Each team member is allowed to use one computer.

1.5.5 Software resources:

It is expected that the teams participating in this competition have adequate knowledge in choosing the best tools to solve the given problem. The MEC will not place any restrictions on what languages are to be used for the competition.

1.5.6 Plagiarism:

Since the use of the Internet and other external resources is permitted in this competition, all information used by competitors must be referenced very carefully. Competitors are not permitted to submit work completed by anyone other than the members of their team. If they decide to recycle their own or someone else's code, it must be clearly cited in the presentation. In addition, the competitors also need to clearly explain why and where the recycled code was used in their software. The judges have the right to ask any team member to describe what a particular section of the code does at any given point during the presentation. If there is any evidence that competitors are

submitting plagiarized work, the entire team will be eliminated from the competition. Competitors are expected to act in good faith with the spirit of the competition.

Generative AI tools may be used for permitted support activities, provided their use is properly cited and the submitted work reflects the team's own technical understanding. Competitors must review and meaningfully modify AI-generated material used in their solution. Submitting complete, unmodified AI-generated solutions or presenting another source's work as original is prohibited.

1.6 PROCEDURE/TIMELINE:

1.6.1 Pre-Competition:

Prior to the competition, information about the rules and procedures will be communicated to registered competitors through official MEC communication channels.

1.6.2 Competition:

1.6.2.1 Problem presentation:

The problem must be presented to all competitors and judges at the beginning of the competition. The competition coordinator(s) must provide detailed explanations of what is expected from the competitors, both orally and in writing.

1.6.2.2 Question period:

Competitors then have fifteen (15) minutes to ask the competition director(s) any questions. Only the time used to ask the questions (not the answers) should be counted in the fifteen (15) minutes. Time used by judges to pose questions will not be counted.

1.6.2.3 Solution development:

Teams will be given eight (8) hours to develop their solutions, produce all required deliverables, and prepare their presentations. All required deliverables must be submitted before the end of the designated development period, unless otherwise instructed by the Competition Coordinators.

1.6.2.4 Rest period:

Competitors must be allowed a minimum of one (1) hour to rest before the presentation phase starts.

1.6.2.5 Solution Presentations:

Competitors will have a maximum of ten (10) minutes to present and demonstrate their solutions. All team members must be present and participate in the presentation or be penalized by the judges. Judges then have a maximum of five (5) minutes to ask questions. Judges can ask a question at any time during the presentation. The clock must be stopped during these interruptions. Once the development period has ended, teams may not modify their submitted

code or presentation materials. During judging, teams may run and demonstrate their submitted programs using the competition datasets and explain the results generated by their systems.

1.6.3 Timekeeping:

The following rules will be adhered to with respect to timekeeping.

1.6.3.1 During the design phase:

- Time starts when the teams arrive at their respective workstations.
- The time remaining in the competition must be announced at three (3) hours, one (1) hour, thirty (30) minutes and ten (10) minutes before the end of the competition.

1.6.3.2 During the presentation:

- If a judge interrupts the prepared presentation with a question, the presentation timer will be paused during the question and response. The separate five (5) minute question-and-answer period will be timed continuously unless the Competition Coordinators authorize a pause.
- The remaining presentation time must be indicated to competitors five (5) minutes and one (1) minute before the end of the allotted presentation period.
- A countdown presentation that is visible to both the judges and presenters must be given during the last 30 seconds of the presentation.

1.6.4 Presentation order:

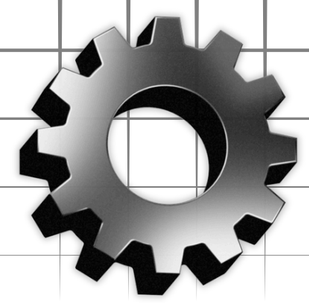
- Presentation order will be determined randomly.
- Presentation order will be announced thirty (30) minutes before the presentations commence. All teams are required to be present at this announcement.
- Teams are not allowed to switch places in the presentation order.
- Teams may not be in the audience for presentations of their competition until after they have completed their presentation. Any attempts at entering the presentation room before their allotted time will result in a warning for a first offence, and a ten (10) point penalty for any further offence.

1.6.5 Deliverables:

1.6.5.1 For competitors:

Prior to the competition, the competition rules, abstract, and relevant information will be communicated to registered competitors through official MEC communication channels.

At the competition briefing, teams will receive access to the official competition materials, including the problem statement, datasets, submission requirements, and judging information. Competitors must follow the access and submission instructions provided by the Competition Coordinators.



1.6.5.2 From competitors:

Each team is required to electronically submit all deliverables outlined by the problem before the end of the design period. This must include the code which will be used to evaluate the team's solution and their presentation. Only the code provided to the Programming Coordinator(s) will be used for evaluation.

Submitted code must include the files and instructions necessary for the Competition Coordinators and judges to run and evaluate the solution. Any required dependencies, setup procedures, and execution commands must be clearly documented.

1.6.6 Response to Questions:

- Only the competition coordinator(s) may answer questions during the problem presentation and design phases. Volunteers and judges may not answer questions regarding rules and procedures at any time.
- During the presentation of the problem, the competition coordinator(s) shall answer orally and write down the answers provided.
- During the design phase, only questions related to deliverable content shall be answered. No answers shall be provided in response to questions about the problem that might lead to the development of a new approach or which might invalidate a solution.
- During the design phase, answers to questions shall be provided in writing to all teams at the same time. Fifteen (15) minutes may be allowed to elapse for a significant number of questions to accumulate before responses are provided. A copy of the responses must be provided to the judges before the presentations.

1.7 ASSESSMENT AND JUDGING:

- The panel must have an odd number of judges and at least three (3) judges.
- The presentations shall be carried out without an audience.
- Feedback forms shall be provided to each team following the announcement of winners, but before the end of MEC.

1.7.1 Judging Matrix:

Competitors will be evaluated using the official Programming Competition Scorecard provided by the Competition Organizers. The scorecard will outline the judging categories, evaluation criteria, and

maximum points available for each criterion. The total score will be calculated out of one hundred (100) points before any applicable deductions.

Category	Judging Criteria	Points
Core Algorithms & Data Structures	A. Routing: Evaluates the correctness, efficiency, and adaptability of the routing algorithm, including its ability to process network changes and recalculate routes when necessary.	/15
	B. Priority Queue: Evaluates the implementation of an appropriate priority queue, correct prioritization logic, and efficient processing of requests or tasks.	/10
	C. Resource Matching: Assesses the effectiveness of resource allocation, consideration of availability and constraints, and the suitability of matching decisions.	/15
Core Algorithms & Data Structures Subtotal		/40
System Design: Inputs, Scoring & State Machine	A. Crisis Input & Scoring: Evaluates input validation, data processing, and the consistency and justification of the system's scoring or prioritization methodology.	/15
	B. Crisis State Machine: Evaluates the correctness of state transitions, handling of changing conditions, and consistency of system behaviour.	/10
	C. Real-Time Communication: Assesses the implementation of event-driven communication, information updates, and coordination between system components.	/10
System Design Subtotal		/35

Dashboard, Constraints & Deliverables	A. Optional Live Website Dashboard: Evaluates the functionality, usability, and clarity of an optional live website dashboard for displaying system information and results.	/5
	B. Constraints & Hidden Test Cases: Evaluates the program's ability to handle technical constraints, unexpected inputs, edge cases, and additional testing scenarios.	/10
	C. Expected Output & Engineering Decisions: Evaluates the completeness and correctness of outputs, required deliverables, and the team's ability to justify its technical decisions.	/10
Dashboard, Constraints & Deliverables Subtotal		/25
TOTAL		/100

1.7.2 Penalty Matrix:

Point Penalties	Confirmed plagiarism or deliberate misrepresentation of another person's work as original	Disqualification
	Insufficient Citation	-50
	Documents Received After Deadline	-50
	Absent Team Member	-25
	Entering the presentation room before the assigned time without authorization	First offence: warning; each subsequent offence: -10
Total		